Homage Programming Guidebook

A practical literacy map for the real-world phone app, Document Studio, local-first data, and backend conventions.

Project root: /Users/davidwilliamclark/projects/homage

Current focus: phone-first Document Studio. Sky, music, quest, and desktop-facing areas are intentionally parked for now.

How To Read This Guide

This guide explains the concepts you need to be literate in the code that already exists in Homage. It is not a complete programming textbook. It is a map of the languages, frameworks, conventions, and mental models that appear in the current app.

- Frontend code lives in app/. It is an Expo React Native app using TypeScript and Expo Router.
- Route screens live in app/app/. This double app/app path is correct: the first app is the Expo project, the second app is the router directory.
- Shared frontend code lives in app/lib, app/hooks, app/components, and app/sky.
- Backend code lives in backend/. It is a FastAPI app with Pydantic request models and a SQLite local-first database.
- The current product focus is Document Studio: write content on the phone, validate safety/location, generate LaTeX/PDF on the backend, and save the result as a local artifact.

1. Architecture At A Glance

Concept	What it means in Homage
Expo React Native	The phone app framework. It lets one TypeScript/React codebase run in Expo Go, iOS, Android, and web preview.
Expo Router	File-based navigation. A file like app/app/document-studio.tsx becomes the /document-studio route.
FastAPI	The Python backend framework. Routes such as POST /documents and GET /local/database are declared with decorators.
SQLite	The local-first metadata store on the backend. It tracks accounts, devices, players, quests, artifacts, pairing codes, and sync events.

Concept	What it means in Homage
Files on disk	Large generated assets stay as files: PDFs, LaTeX source, uploaded photos, audio, and song JSON.
AsyncStorage	A small key-value store on the phone used for local profile/session/artifact metadata.
Expo FileSystem	Phone-side persistent file storage used to keep downloaded PDFs, LaTeX source, songs, and images.

Current data loop

1. The phone screen gathers user input and local state with React hooks.
2. The API helper serializes a typed payload and sends it to FastAPI.
3. FastAPI validates it with Pydantic, writes files, records metadata in SQLite, and returns URLs plus structured JSON.
4. The phone downloads durable artifacts into Expo FileSystem and records an index entry in AsyncStorage.
5. Admin and library screens read from backend status endpoints and local artifact helpers to show what exists.

2. TypeScript And React Native Literacy

Most frontend files are TypeScript React components. The important thing to recognize is that UI, state, and side effects are expressed as functions rather than classes.

Concept	What it means in Homage
TSX	TypeScript plus JSX. Files ending in .tsx can contain typed JavaScript and React UI markup.
Component	A function that returns UI. Example: DocumentStudioScreen returns SafeAreaView, ScrollView, TextInput, Pressable, and Text elements.
Props	Inputs passed to components. Local helper components often receive selected values, options, and callbacks.
useState	Stores local screen state, such as title, body, template, publishing, error, and success.
useEffect	Runs side effects such as loading profile data or resetting safe-area confirmation when GPS readiness changes.
useMemo	Caches derived values, such as preview colors selected from the template.
Async function	A function that returns a Promise and can await network/file operations.
Union type	A fixed set of legal values, like 'poem' | 'prose_poem' | 'short_story'. These mirror backend validation rules.

Pattern to recognize in screens

A Homage screen usually follows this rhythm:

```
imports -> type definitions -> option constants -> component state -> effects -> derived values ->
event handlers -> loading state -> returned UI -> styles
```

When reading a screen, find the event handlers first. For Document Studio, publish() is the key handler because it reveals the entire product loop: form validation, API call, backend response, artifact save, success/error state.

3. Expo Router And App Structure

Expo Router turns files into routes. The route folder is app/app because the Expo project itself is app. Do not move route files to the repository-level app folder.

Concept	What it means in Homage
app/app/index.tsx	The home screen. It now points users toward Document Studio, Admin, Pair Device, Terminal Help, and Artifacts.
app/app/_layout.tsx	The shared Stack navigator configuration: titles, header styling, background styling.
app/app/document-studio.tsx	The active writing and publishing screen.
app/app/admin.tsx	A dashboard for backend status, SQLite status, pairing, and latest documents.
app/app/artifacts.tsx	The local artifact library screen.
app/app/terminal.tsx	An allowlisted help/admin console, not a real shell.

Navigation convention

Use useRouter() from expo-router and router.push('/route') to navigate. The Href type helps keep route strings disciplined. For simple links in code, prefer explicit route names over clever dynamic strings.

4. Styling And UI Conventions

React Native styling is written with StyleSheet.create. It resembles CSS, but property names use camelCase and layout uses Flexbox by default.

Concept	What it means in Homage
StyleSheet.create	Creates named style objects. Screens keep styles at the bottom of the file.
RGB_THEME	Shared palette in app/lib/theme.ts. Use it instead of scattering raw colors.
Pressable	Touchable UI element for buttons and rows.
TextInput	Native input field. Document Studio uses maxLength, placeholderTextColor, selectionColor, and controlled value state.
SafeAreaView	Avoids notches and system UI overlap on phones.
KeyboardAvoidingView	Improves typing ergonomics when the keyboard appears.
ActivityIndicator	Spinner for loading or submitting states.

For phone-first work, pay attention to keyboard behavior, scroll behavior, loading states, disabled states, and error messages. These matter more than desktop polish right now.

5. API Contracts And Shared Types

The frontend API helpers in app/api.ts define what the phone believes the backend accepts and returns. The backend Pydantic models in backend/server.py define what the server actually accepts. Keeping those aligned is one of the most important maintenance tasks.

Concept	What it means in Homage
API_BASE	Resolved from Expo config extra values, with a production fallback. This controls which backend the phone talks to.
fetch	The browser/React Native network primitive used by helpers like createDocument and getDocuments.
JSON.stringify	Turns TypeScript objects into JSON request bodies.
res.ok	A success check. Helpers throw meaningful errors when backend calls fail.
Pydantic Field	Backend validation for length, ranges, regex patterns, and required fields.
Typed response	Frontend functions promise shapes such as PublishedDocument or LocalDatabaseStatus.

Document payload alignment

Document Studio sends fields such as player_id, title, body, document_kind, template, accent, typography, alignment, puzzle_kind, puzzle_prompt, puzzle_answer, latitude, longitude, accuracy, stationary_fixes, and safe_area_confirmed. The backend DocumentCreateRequest validates each of these before doing any work.

6. FastAPI And Python Backend Literacy

FastAPI uses decorators to bind Python functions to HTTP endpoints. Pydantic models sit at the boundary and reject invalid data before the endpoint body runs.

Concept	What it means in Homage
@app.post('/documents')	Declares the create document endpoint.
BaseModel	Pydantic class used to define request body schemas.
Field	Validation rule: min_length, max_length, numeric bounds, or regex pattern.
HTTPException	A structured API error with a status code and detail message.
FileResponse	Sends generated PDF, LaTeX, song, or upload files back to the client.
lifespan	Startup/shutdown hook. Homage creates data directories and initializes SQLite during startup.
CORS	Configured permissively for prototype/dev access from app and web preview.

Backend helper conventions

- Keep path validation strict before serving files. document_path checks prefixes, characters, and allowed extensions.
- Normalize user-facing text before hashing or typesetting. puzzle answers and LaTeX strings are cleaned deliberately.
- Use explicit records and JSON payloads for storage rather than hidden object mutation.
- When returning public records, strip private fields such as created_by, source_path, pdf_path, and answer_digest.

7. SQLite And Local-First Data

The project direction is local-first: SQLite is the canonical metadata/event store, while large assets remain files. In practice this means app behavior should be understandable even when the network or a cloud backend is not the center of the design.

Concept	What it means in Homage
schema_meta	Stores schema version information.
local_accounts	Represents a local household/base account.
devices	Represents phones or other devices paired to an account.
players	Represents app identities used in multiplayer or publishing flows.
quests	Stores generated or community quest metadata.
artifacts	Stores document, song, and photo artifact metadata.
sync_events	Append-only event log for things like device_paired and document_created.
pairing_codes	Short-lived codes used to link a phone to a local account.
WAL mode	SQLite journal mode that improves concurrent read/write behavior.

Mental model

SQLite answers 'what happened and what exists?' File storage answers 'where is the big object?' A document record points to a PDF URL and source URL; the PDF and .tex file are not shoved into SQLite.

8. Document Studio End-To-End

1. The user opens /document-studio and waits for profile and location readiness.
2. The screen gathers title, byline, body, kind, template, accent, typography, alignment, puzzle prompt, and puzzle answer.
3. useStationaryAuthoringLocation requires accurate GPS, limited speed, limited drift, and three accepted fixes.
4. The user confirms the safe area. This is separate from GPS because safety is a human decision.
5. publish() calls createDocument(), which POSTs to /documents.

6 FastAPI validates the payload, creates a document id, writes a .tex source file, compiles a PDF, writes JSON metadata, creates a quest-like record, records SQLite metadata, and appends a sync event.

7 The phone receives a PublishedDocument and saves the PDF/source as a local Artifact with expo-file-system and AsyncStorage.

LaTeX literacy

Concept	What it means in Homage
Escaping	latex_escape prevents user text from breaking the .tex file when it contains characters like %, _, &, $, or braces.
Templates	neon, terminal, and letterpress map to background, foreground, and muted colors.
Typography	serif, sans, and mono map to LaTeX font commands.
Alignment	left, center, and right map to LaTeX environments.
PDF compiler	The backend currently expects a LaTeX compiler to turn .tex into .pdf. Your local Org workflow uses tectonic.

9. Phone Capabilities And Safety

Homage uses phone capabilities as product material: location, camera, file storage, sensors, and links. That means permissions and conservative safety policy are part of the app architecture, not an afterthought.

Concept	What it means in Homage
expo-location	Requests foreground permission and samples GPS for authoring and exploration.
Accuracy threshold	Document authoring requires accuracy at or below 35 meters.
Stationary fixes	The hook requires three accepted fixes and rejects high speed or large drift.
safe_area_confirmed	A required boolean for publishing location-tied content.
Backend authority	Quest safety policy remains server-side. Client fallbacks should stay conservative.
Tunnel testing	Expo tunnel allows remote phone testing through Expo Go. Node 20 worked for this project when Node 25 broke the tunnel wrapper.

10. Files, Storage, And Artifacts

Artifacts are the bridge between generated media and the user-visible library. They keep enough metadata to browse and reopen things later, while the actual files live in persistent phone storage or backend data directories.

Concept	What it means in Homage
AsyncStorage key	homage.artifacts.v1 stores the local artifact index.
MAX_ARTIFACTS	The local index is capped at 200 entries.

Concept	What it means in Homage
FileSystem.documentDirectory	The phone's durable document storage root.
documents/	Phone folder for saved PDF and .tex files.
songs/	Phone folder for rendered song audio.
artifacts/	Phone folder for persisted images.
Download headers	ngrok-skip-browser-warning is added when pulling files through ngrok-style URLs.

Important convention: artifact save functions check whether an item already exists before adding a duplicate. This keeps repeated opens from filling the library with copies.

11. Tests, Tooling, And Known Rough Edges

Concept	What it means in Homage
npx tsc --noEmit	Type-checks the Expo app without building. Current known unrelated failures are in app/maze.tsx and state/quests.ts.
Expo package warnings	Expo SDK patch versions are behind expected compatible patches. This is a maintenance item, not necessarily an immediate blocker.
backend/test_*.py	Backend tests cover local database, multiplayer, and quest behavior.
npx expo start --tunnel -c	Phone-focused remote launch command. Use Node 20 if tunnel fails under newer Node.
Org planning	Your planning template and proposal files live under planning/ and can export to PDF with tectonic.

Practical debugging sequence

1. Read the exact error, including file path and endpoint.
2. Identify whether it is frontend state/UI, API contract, backend validation, file generation, SQLite persistence, or phone permission/storage.
3. Check the type definitions on both sides of the boundary.
4. Make the smallest change that matches the existing local pattern.
5. Run the narrowest useful check first, then broader checks if the change touched shared behavior.

12. Glossary For Becoming Literate

Concept	What it means in Homage
API	A contract between pieces of software. In Homage, the phone calls FastAPI endpoints.
Backend	The Python server that validates requests, writes files, and records SQLite metadata.

Concept	What it means in Homage
Client	The phone app.
Endpoint	A URL plus HTTP method handled by FastAPI, such as POST /documents.
Hook	A React function beginning with use, such as useStationaryAuthoringLocation.
Local-first	Design where local data and files are primary, not just a cache of a remote cloud.
Metadata	Small structured facts about an object: title, id, owner, URL, timestamps, location.
Payload	The JSON object sent to or returned from an API.
Route	A screen path in Expo Router.
Schema	The shape of data in a database or request model.
State	Values a screen remembers while it is open.
Type	A TypeScript or Pydantic description of what values are allowed.

13. What To Learn Next

You do not need to learn everything at once. For the current Document Studio phase, this is the highest-leverage order:

1. React component state: useState, controlled TextInput, conditional rendering, and disabled buttons.
2. Expo Router routes: how file paths become phone screens.
3. TypeScript unions and object types: how the app restricts values like template and document_kind.
4. Fetch and API helpers: how phone actions become backend requests.
5. Pydantic validation: how the backend protects itself from malformed input.
6. SQLite tables and JSON payloads: how local-first metadata is recorded.
7. LaTeX generation: how user writing becomes a durable PDF artifact.
8. Phone permissions and file storage: how location, safety, and persistent documents work.

The north star: understand the full path from a typed word on the phone to a saved PDF artifact with a SQLite record. Once that loop feels clear, the rest of Homage becomes much easier to reason about.